Project Design Phase

Solution Architecture

Date	September 2026
Project Name	ServiceNow-Based Employee Laptop Request Application

Solution Architecture:

The solution architecture defines how the ServiceNow-based employee laptop request application connects users, request workflows, approvals, asset information, notifications, and reporting. The architecture is designed to provide a centralized and transparent process for submitting, approving, tracking, and fulfilling laptop requests.

- Provide employees with a simple interface to submit laptop requests and view request status.
- Use ServiceNow workflow and approval rules to route requests to the appropriate managers or approvers.
- Validate request information and maintain request, employee, and asset records in ServiceNow.
- Check laptop availability and support assignment of an available device to the employee.
- Send notifications for request submission, approval decisions, status changes, and fulfillment updates.
- Provide role-based access and reporting so employees, managers, and IT teams see the information relevant to them.

Project Solution Architecture Diagram:

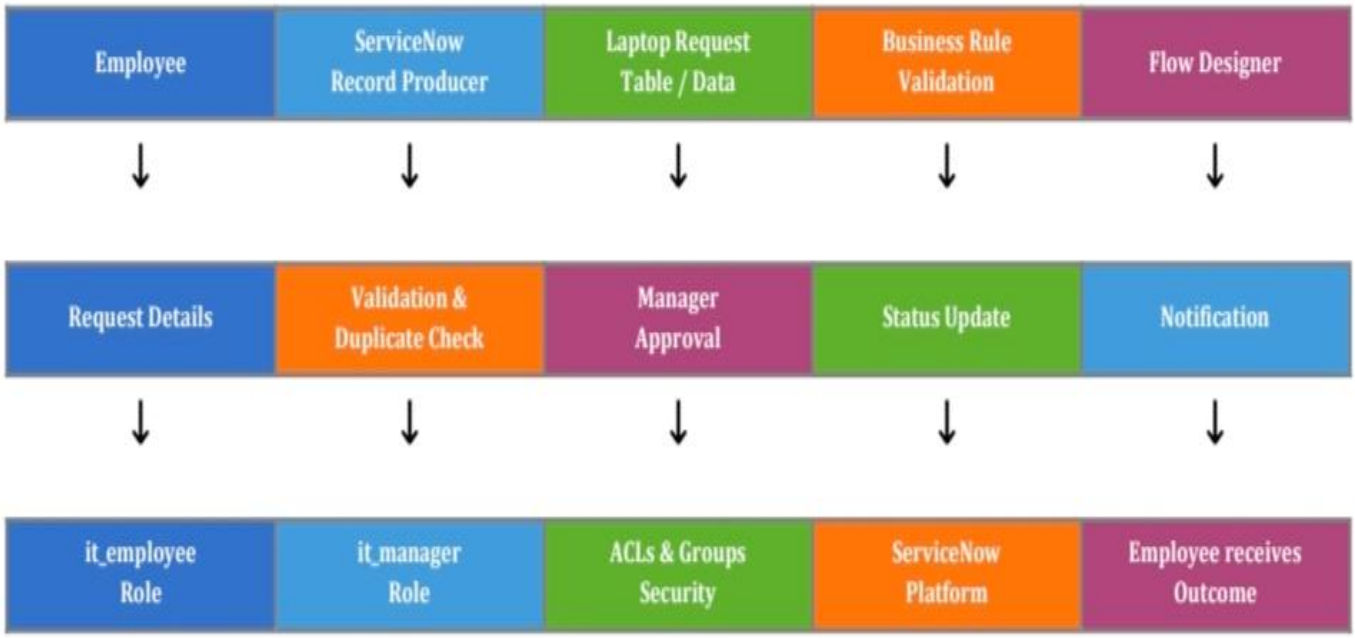


Figure 1: Solution architecture and information flow for the ServiceNow-based employee laptop request application.

Architecture Components:

Component	Purpose in the Project	Technology / Platform
User Interface	Employee request form, request history, and status view.	ServiceNow Service Portal / UI
Application Logic	Request validation, workflow routing, approval rules, and status updates.	ServiceNow Workflows / Flow Designer / Business Rules
Request Data	Stores laptop requests, approvals, comments, and request status.	ServiceNow tables / database
Asset & CMDB	Checks laptop availability and supports asset assignment and tracking.	ServiceNow Asset Management / CMDB
Notifications	Communicates submission, approval, rejection, and fulfillment updates.	ServiceNow Notifications / Email
Security	Controls access based on employee, manager, and IT roles.	ServiceNow Roles / ACLs
Reporting	Provides request, approval, fulfillment, and status reports.	ServiceNow Reports / Dashboards